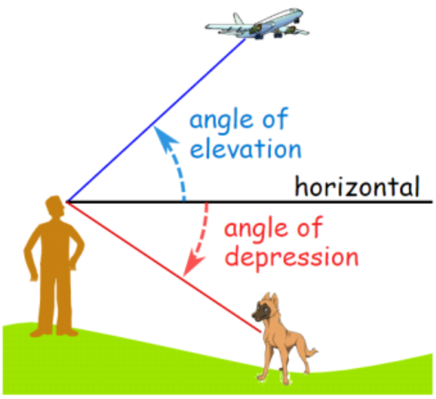
Week 4 - Lesson 2 - Angles of Elevation and Depression

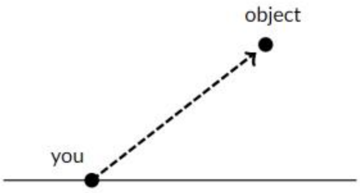
Thursday, 9 April 2020 11:36 AM

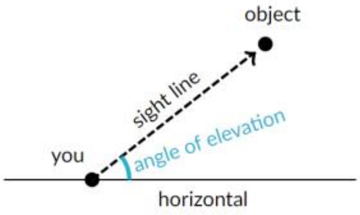
The "upwards" angle from the horizontal to a line of sight from the observer to some point of interest.

If the angle goes "downwards" it is called an Angle of Depression.

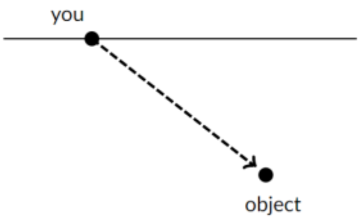


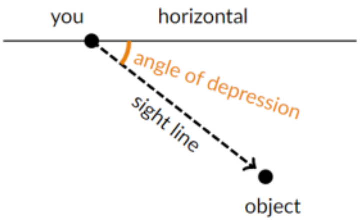
When you see an object above you, there's an angle of elevation between the horizontal and your line of sight to the object.





Similarly, when you see an object below you, there's an angle of depression between the horizontal and your line of sight to the object.

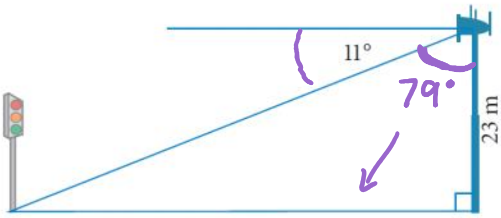




Example 1

In this exercise, express your answers correct to 1 decimal place (for lengths) or the nearest degree (for angles), unless instructed otherwise.

1 From the top of a 23 m high mobile phone tower, the angle of depression to the bottom of a set of traffic lights is 11°. How far is the set of traffic lights from the base of the mobile phone tower?



79°

A

x O

90 − 11
= 79

Usually the angle of depression is NOT the top angle in the triangle. Subtract the angle of depression from 90° to determine the size of the top angle.

SOH CAH TOA

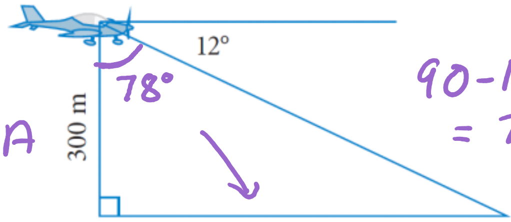
$$23 \times \tan 79° = \frac{x}{\cancel{23}} \times \cancel{23}$$

$$x = 23 \times \tan 79°$$

$$x = 118.3\text{m}$$

Example 2

2 Samantha is flying a light aircraft at a height of 300 m above the ground. Her angle of depression to the landing strip is 12°. Calculate the horizontal distance from the ultralight to the landing strip.



78°

A

x
O

90 − 12
= 78°

$$300 \times \tan 78° = \frac{x}{\cancel{300}} \times \cancel{300}$$

$$x = 300 \times \tan 78°$$

$$x = 1,411.4\text{m}$$